

SERVICE MANUAL

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0273S

Revision 3

GE 1.5T HD TR Knee Array

GE Part Number: M3087JF

MRIDC Part Number: 800273



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Rev. June 13, 2006

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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
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FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins ááanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnaða eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

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(JA)

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- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者

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경고
(KO)

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- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS
(LV)

Šī apkopes rokasgrāmata ir pieejama tikai anglu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumų risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS
(LT)

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL
(NO)

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE
(PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT
(TR)

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE 1.5T HD TR Knee Array.



Note: Coils do not ship with phantoms. Phantoms come in a unified phantom set with the MR System.

SHIPPING LIST – TABLE 1-1

Description	GE Part #	Qty
1.5T HD TR Knee Array	5114356-2	1
Operator Manual	5114356-5	1
Service Manual	5114356-6	1
Thin Knee Pad	5114356-7	1
Thick Knee Pad	5114356-8	1
Ramp Pad	5114356-9	1
Ankle Pad	5114356-10	1
Phantom Positioner	5114356-12	1

1-2 Compatibility

This coil is compatible with the following 1.5T hardware configurations:

Signa Excite HD, HDx, HDe, Signa HDxT, Discovery MR450, and Optima MR450w systems

1-3 Related Documentation

- 5133315 - Signa EXCITE HD 1.5T 11.x Service Methods
- 5155500 – Signa HDx 14.x Service Methods
- 5323683-2EN – Excite HDe Service Methods
- 5440000 – Signa 15.x HDxT Service Methods

- 5500110 - Discovery 20.x MR450 1.5T Service Methods
- 5670000 – Optima MR450w 1.5T Service Methods
- 1.5T HD TR Knee Array Operator's Manual, 5114356-5

1-4 Environmental Requirements

Storage

Store the coil in an air-conditioned scan room or equipment room. To store the coil, a storage space of 45.7 cm/18.0 in (width) x 72.4 cm/28.5 in (depth) x 42.7 cm/16.8 in (height) is required.

Dimensions

Coil:	45.7 cm x 72.4 cm x 42.7 cm	(18.0 in x 28.5 in x 16.8 in)
Phantom Positioner:	17.8 cm x 17.8 cm x 19.9 cm	(7.0 in x 7.0 in x 7.83 in)

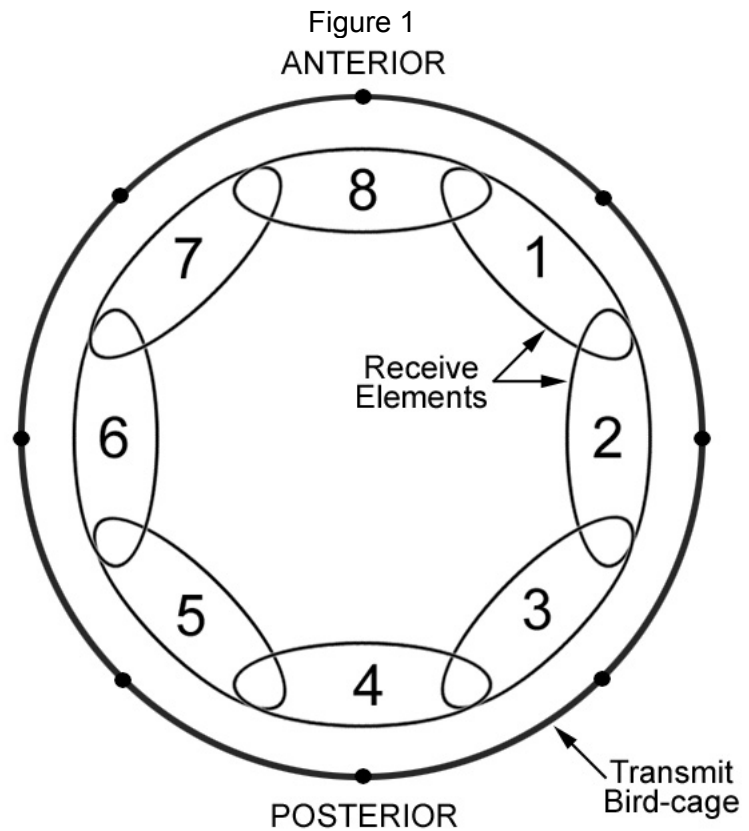
Weight

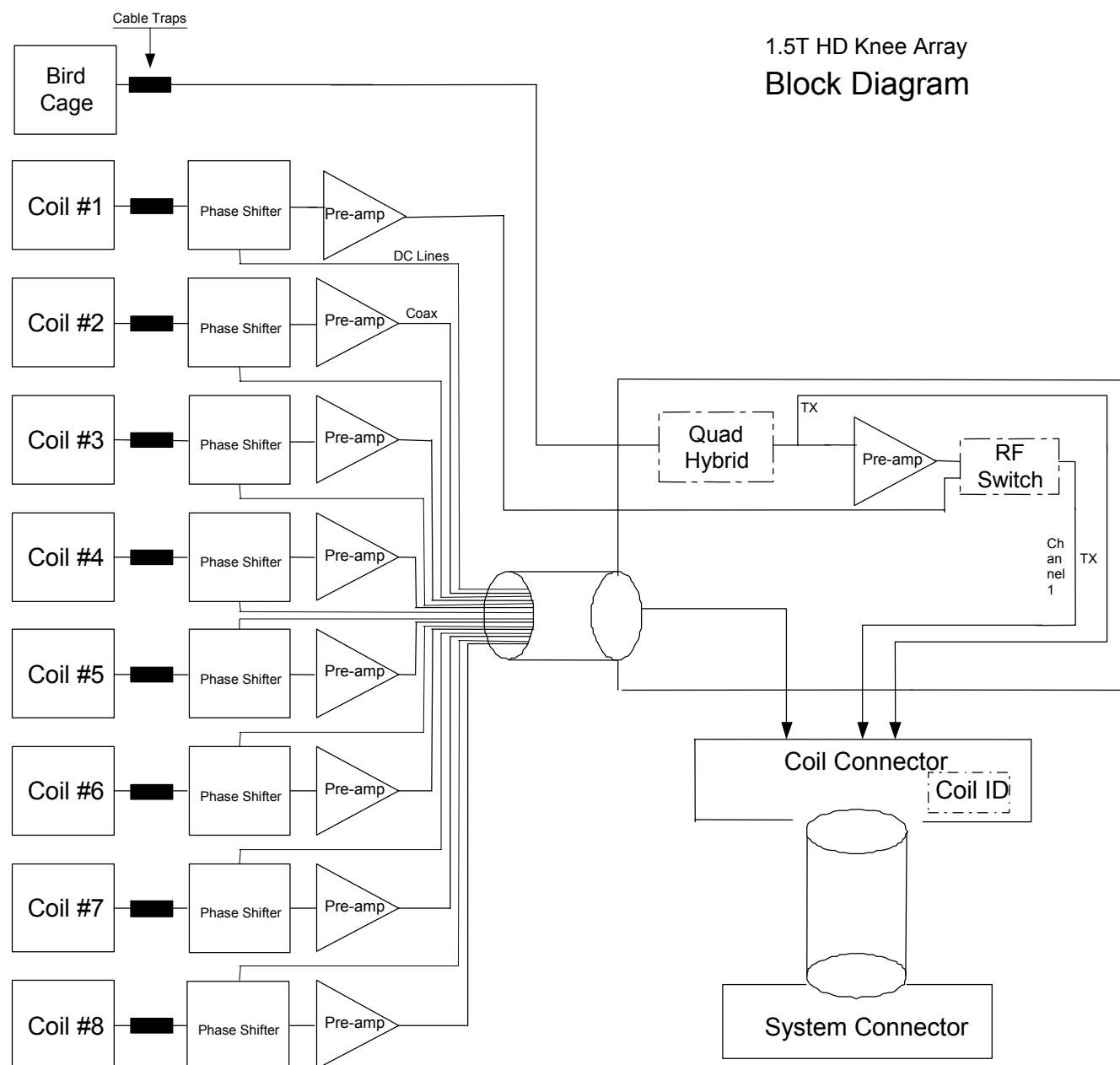
Coil:	8.56 Kg	(18.9 lb)
Phantom Positioner:	1.5 Kg	(3.3 lb)

1-5 Theory of Operation

Refer to Figure 1 below and the Block Diagram on the following page.

The 1.5T HD TR Knee Array consists of 8 receive elements, and a quad birdcage transmit coil. Each receive element is comprised of a loop arranged azimuthally around the knee. Each loop is approximately 45° in azimuthal extent. The center of Loop #1 is located 45° rotated in the clockwise direction (as viewed from the head) with respect to the center of the patella. Each of the next seven loops is rotated clockwise by 45° successively. During transmit, each receive loop is decoupled with both passive and active decoupling, with an additional active preamp protection network. During receive, the transmit bird-cage is decoupled and signals from elements 1 – 8 are directed to separate ultra-low-input impedance preamp modules placed directly at the output of the loop. These preamp modules provide additional element decoupling. In addition, the 1.5T HD TR Knee Array allows receiving with the bird-cage coil to obtain a uniform signal from the imaging volume.





SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Reverse Polarity Configuration



DURING REVERSE POLARITY CONFIGURATION, THE 1.5T HD TR KNEE ARRAY COMBINER BOARD IS EXPOSED. **USE ESD SAFETY PRECAUTIONS**

2-1-1-1 Tools Required

1ea Philips Screwdriver

2-1-1-2 Configuration Time

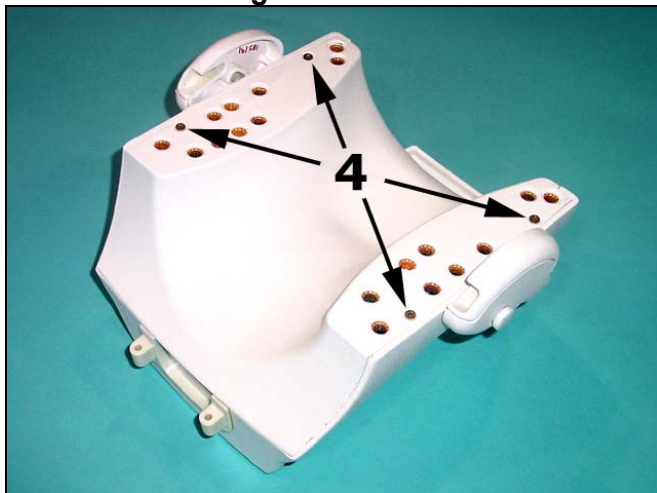
15 to 20 minutes

2-1-1-3 Configuration Procedure

If required, follow these steps to configure the 1.5T HD TR Knee Array for reverse polarity.

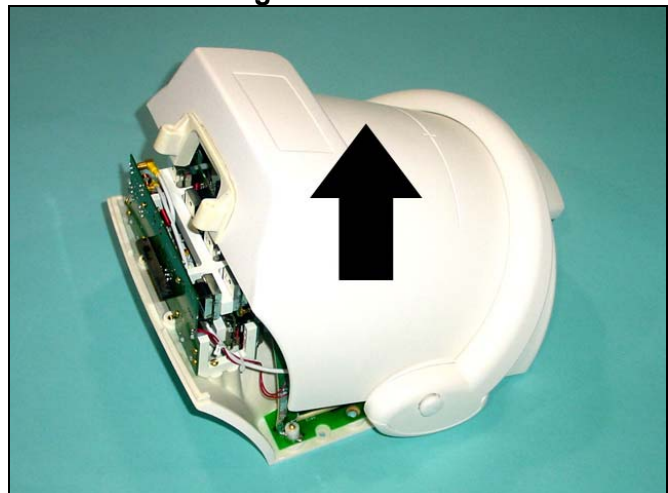
1. Remove the 1.5T HD TR Knee Array coil cable as detailed in section **5-2 External Cable Replacement**.
2. Separate the anterior and posterior coil sections.
3. Using the phillips screw driver, remove 4 screws from the anterior coil section as illustrated in Figure 2-1-1-3-1.
4. Remove the anterior coil section cover as illustrated in Figure 2-1-1-3-2.

Figure 2-1-1-3-1



Remove 4 screws

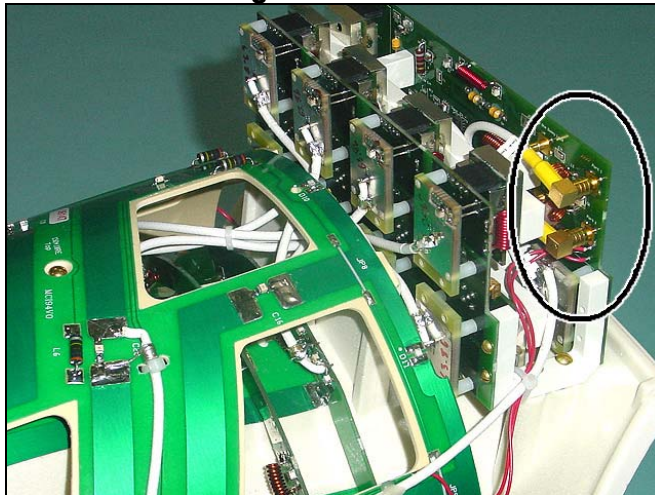
Figure 2-1-1-3-2



Remove Cover

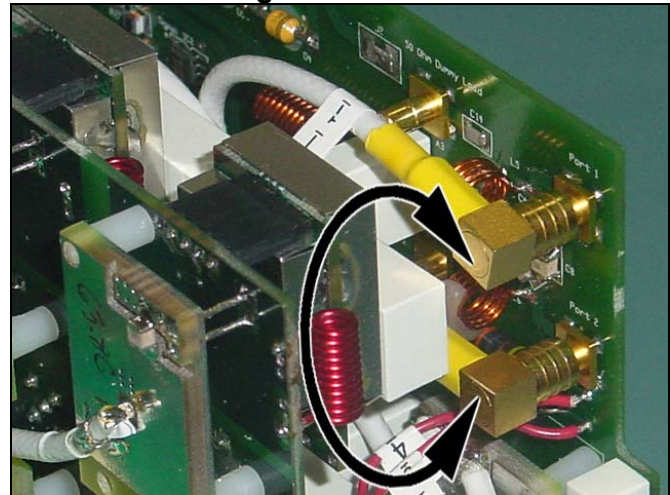
5. Locate the two *Trap/Combiner Coaxial Cables*; they are marked 1 and 2, and are connected to the *Combiner Printed Circuit Board* at locations *Port 1* and *Port 2* respectively. Reference Figure 2-1-1-3-3.
6. Disconnect both *Trap/Combiner Coaxial Cables*. Connect cable 1 to *Port 2* and connect cable 2 to *Port 1*. Reference Figure 2-1-1-3-4 and Table 2-1-1-3.

Figure 2-1-1-3-3



Trap/Combiner Cable Connectors

Figure 2-1-1-3-4



Exchange Connectors

POLARITY CABLE CONFIGURATION - TABLE 2-1-1-3

Trap/Combiner Coaxial Cable	For Normal Polarity	For Reverse Polarity
Number 1	Connect to Port 1	Connect to Port 2
Number 2	Connect to Port 2	Connect to Port 1

7. Reinstall the anterior coil section cover using the 4 associated phillips head screws.
8. Reinstall the 1.5T HD TR Knee Array coil cable as detailed in section **5-2 External Cable Replacement**.
9. Perform the **Coil Imaging Performance Verification, Section 3-2**.

2-1-2 Installing the Coil

There are no installation steps other than plugging in the coil and performing functional checks. Coil configurations will already be on the system software. For Discovery MR450 and Optima MR450w the coil configuration is HD TRKnee PA. There are 3 configurations available for HDx, HDxt and HDe:

HD TRKnee PA
HDTRKneeService
HDTRK1chService

HD TRKnee PA is the only configuration visible in clinical mode.

HDTRKneeService and **HDTRK1chService** are visible in service mode (geservice).
HDTRKneeService and **HDTRK1chService** must be installed for the SNR tool to run.

2-2 Quality Assurance Check

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3 - Coil Imaging Performance Verification.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED USING LEGACY PHANTOM – TABLE 3-2-1

Description	GE Part #	MRIDC #	Qty
Phantom	5114356-11	870005	1
Phantom Positioner	5114356-12	870006	1

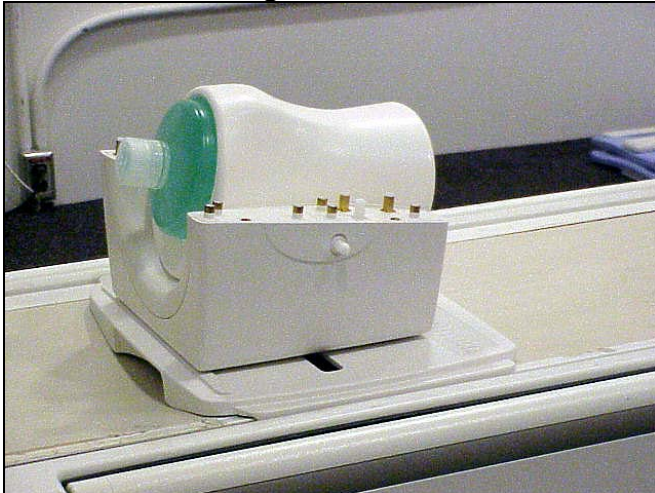
TOOLS REQUIRED USING UNIFIED PHANTOM – TABLE 3-2-2

Description	GE Part #	MRIDC #	Qty
Large cylindrical unified phantom	5342679	N/a	1
Phantom Positioner	5114356-12	870006	1

3-2-2 Data Collection Setup with Legacy Phantom

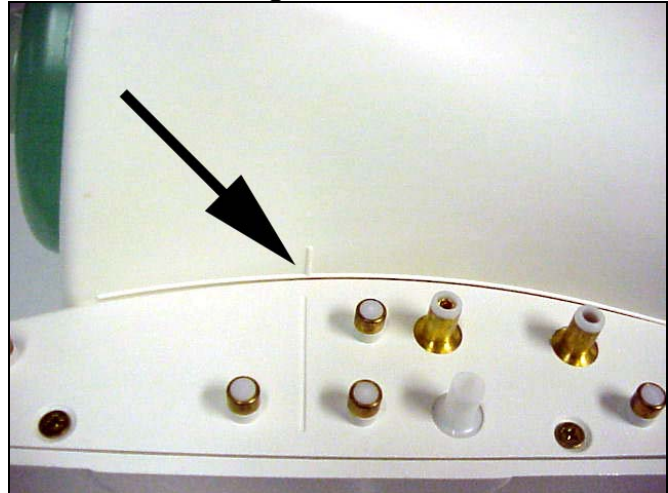
1. Place the 1.5T HD TR Knee Array, the phantom positioner, and the phantom on the patient cradle as shown in Figure 3-2-2-1. Note the serial number of the coil.
2. Ensure that the phantom positioner is aligned with the posterior coil section as indicated by the marks on the positioner. (Figure 3-2-2-2)

Figure 3-2-2-1



Coil and Phantom Setup

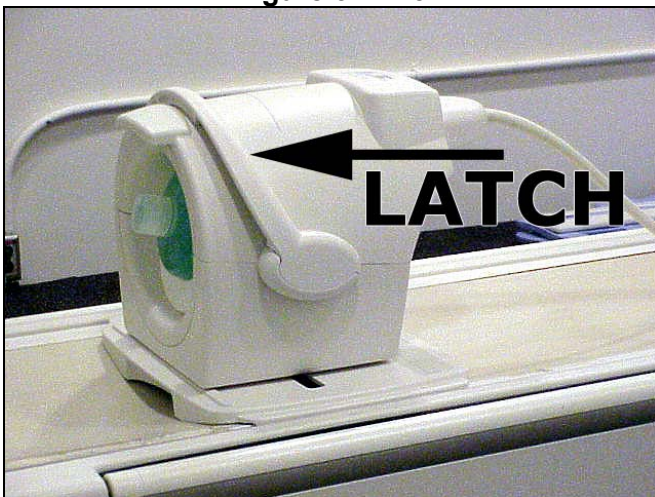
Figure 3-2-2-2



Phantom Positioner Alignment Marks

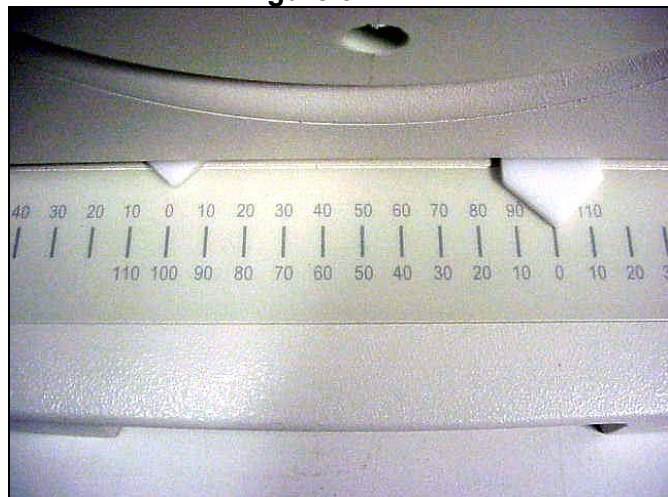
3. Being careful to align the connector pins, place the anterior coil section onto the posterior coil section and latch into place. (Figure 3-2-2-3) The scanner will not operate if the coil sections are not latched correctly.
4. Ensure that the coil is aligned and centered with respect to the baseplate as indicated by the arrows on the baseplate offset scale. (Figure 3-2-2-4)

Figure 3-2-2-3



Anterior Coil Section in Place

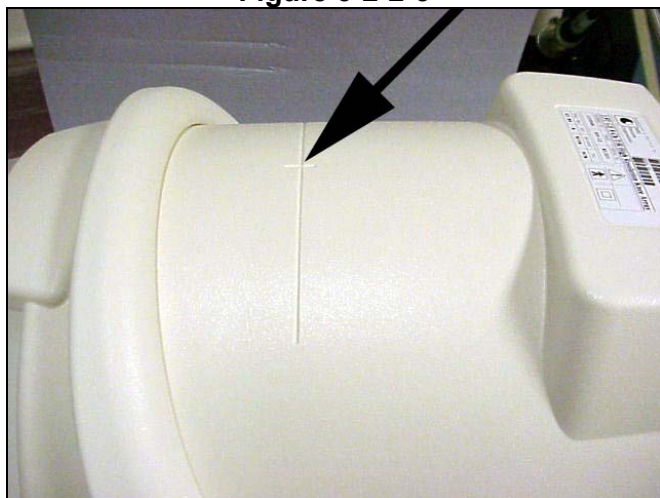
Figure 3-2-2-4



Align and Center the Coil

5. Establish an axial landmark using the reference mark on top of the coil. (Figure 3-2-2-5)
Advance to scan.

Figure 3-2-2-5



Establish Landmark

3-2-3 Data Collection Setup with Unified Phantom

Visual	Description
 Figure 3.2.3.1	1. Place the lower section of the coil on the patient table as shown in Figure 3.2.3.1.
 Figure 3.2.3.2	2. Align the alignment mark on the coil to Zero position on the scale as shown in Figure 3.2.3.2
 Figure 3.2.3.3	3. Place the large cylindrical unified phantom into the Positioner as shown in Figure 3.2.3.3.



Figure 3.2.3.4

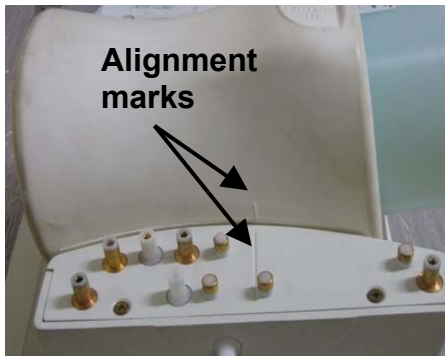


Figure 3.2.3.5

4. Place this Phantom-positioner setup on the lower part of the coil as shown in Figure 3.2.3.4.
5. Ensure that the phantom positioner is aligned with the posterior coil section as indicated by the marks on the positioner and the coil (Figure 3.2.3.5)



Figure 3.2.3.6

6. Carefully align the connector pins on the lower part of the coil with the upper part of the coil and latch both the coil sections into place as in Figure 3.2.3.6. (The scanner will not operate if the coil sections are not latched correctly).

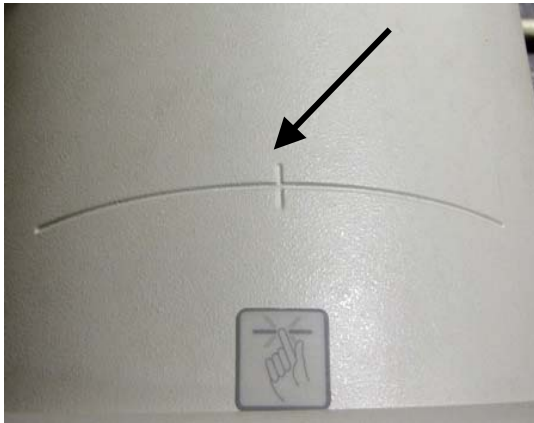


Figure 3.2.3.7

7. Connect the coil to Port A of the system LPCA and Landmark the coil at the cross mark available near the handle of the coil (Figure 3.2.3.7) and advance to scan.

3-2-4 MCQA / SNR Test

1. Move back to the operator workspace and end the previous exam, if required.
2. For 11x/12x systems, go to Common Service Desktop, select Troubleshooting, Image Quality Troubleshooting Tools, and then coil Surface Coils SNR Test. Continue with Step 3 of this section.

For 14x or later systems, go to Common Service Desktop, select Image Quality, and then select MCQA. Steps 3-6 do not apply.

If more information is needed for MCQA, it can be found on 14x/15x service CD or on web at: Troubleshooting/Image Quality/multicoil Quality Assurance. For MR450/450w the path is: Troubleshooting/ System / Multi-Coil Quality Assurance Tool

3. When the Multicoil SNR Tool Control window opens, select the **8 Channel TRKnee** Coil from the Coil Name selection menu. (Figure 3-2-4-1)

Figure 3-2-4-1

The screenshot shows a software window titled "Multicoil SNR Tool Control". At the top is a "File" menu. Below it is a "Scan Panel" with three input fields: "Coil Name" containing "8 Channel TRKnee", "Coil serial #" containing "e1277", and "Passes" containing "1". There are two buttons: "Start Scan" and "Abort Scan". To the right of the "Scan Panel" is a "View Archive Data" button. Below the "Scan Panel" is a "Results" section, which is currently empty and displays the word "EXAMPLE" in large, blue, outlined letters.

Multicoil SNR Tool Control Window

4. Enter the Coil serial number.
5. Enter the number of repetitions in the Passes field. Typically 1 pass is sufficient.
6. Click on the Start Scan button. The system will acquire the images and automatically start data analysis.

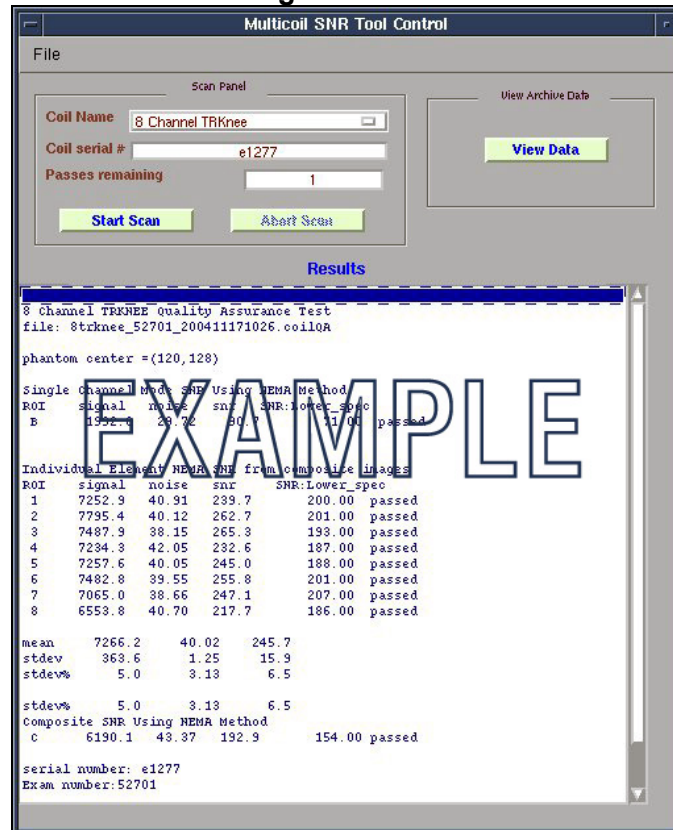
3-2-5 Data Analysis 11.x and 12.x only

For information on MCQA for systems later than 12.x, it can be found on 14x/15x service CD or on web at: Troubleshooting/Image Quality/multicoil Quality Assurance. For MR450/450w the path is: Troubleshooting/ System / Multi-Coil Quality Assurance Tool

The data analysis is automated.

1. When the scan completes, the data will be displayed in the Multicoil SNR Tool Control window. (Figure 3-2-5-1) Each measurement is assigned a pass/fail value. The coil passes the SNR test if all measurements pass.

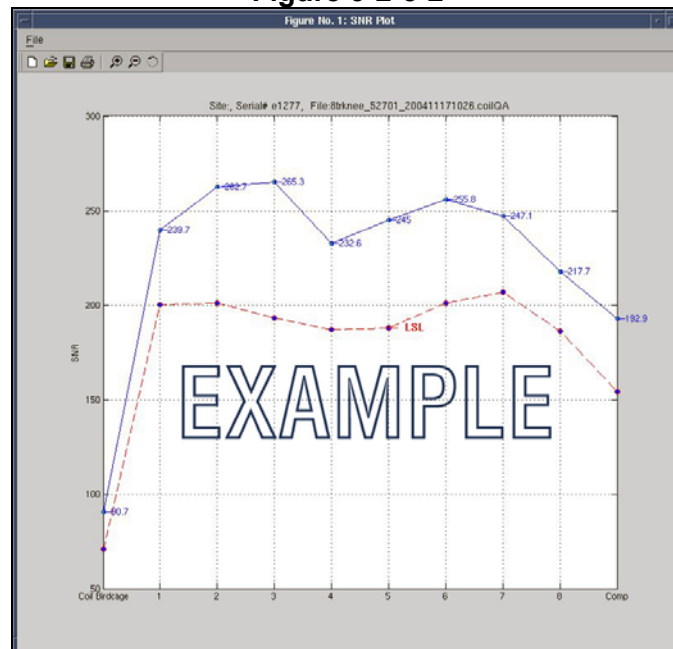
Figure 3-2-5-1



Results in the Multicoil SNR
Tool Control Window

2. A plot window with SNR, indicating the measured value and specification limit, will open. (Figure 3-2-5-2)

Figure 3-2-5-2

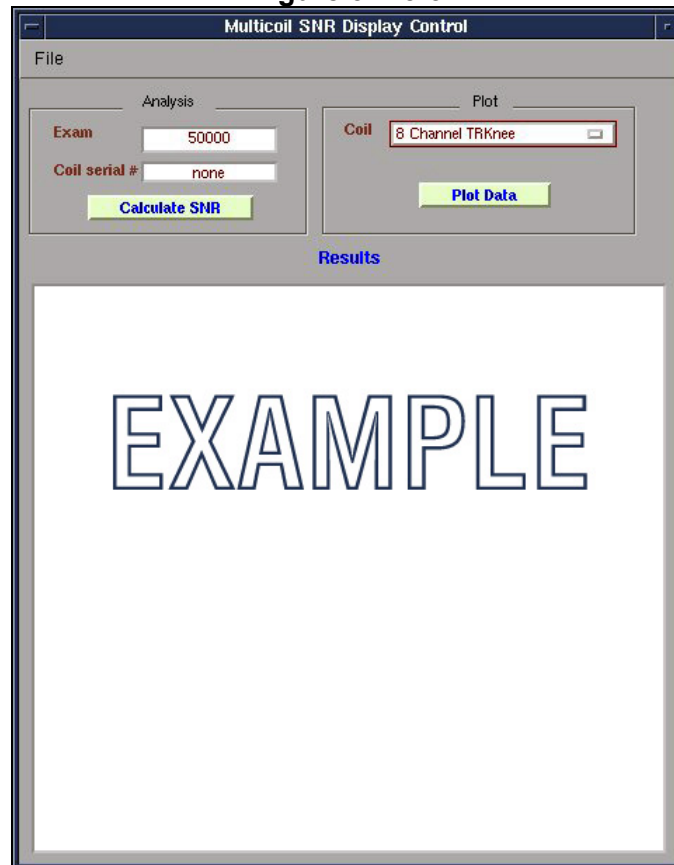


Results in the SNR Plot Window

To view archived data files:

- a) Click on the View Data button in the Multicoil SNR Tool Control window.
(Figure 3-2-2-6)
- b) A new window, Multicoil SNR Display Control, will open. (Figure 3-2-5-3)

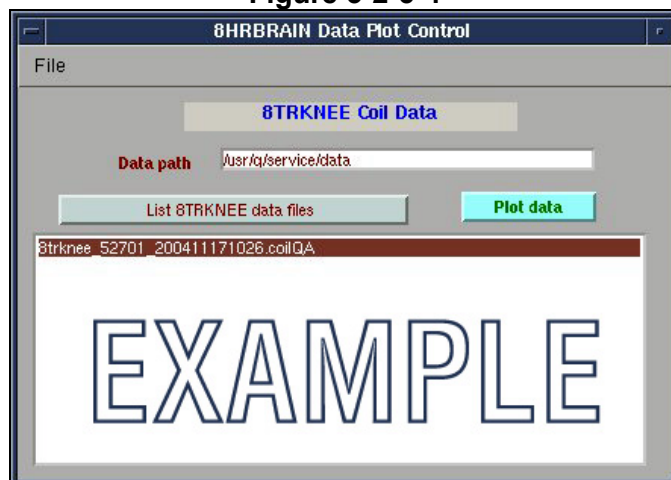
Figure 3-2-5-3



Multicoil SNR Display Control Window

- c) Select the File Tab and turn Expert Mode on.
- d) Select the **8 Channel TRKnee** coil in the coil selection menu.
- e) Click the Plot Data button.
- f) The Data Plot Control window opens. (Figure 3-2-5-4)

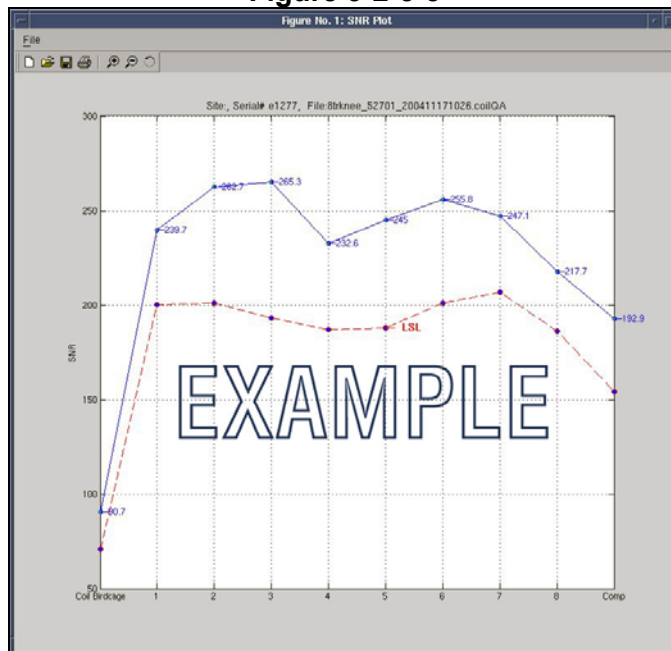
Figure 3-2-5-4



Data Plot Control Window

- g) Select the desired file and click Plot Data.
- h) A plot window with SNR, indicating the measured value and specification limit, will open. (Figure 3-2-5-5)

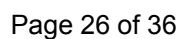
Figure 3-2-5-5



Results in the SNR Plot Window

Use Figure 3-3 for pin identification while performing the procedures in sections 3-3 and 3-4. Referring to Section 5-2, External Cable Replacement, disconnect the coil cable from the coil.

System Connector



3-3-1 RF Signal Channels Check

NOTE: The center pins of the System Connector *receive coaxial connectors* are extremely fragile [Figure 3-3, C1-C4 and D1-D4]. **Do not attempt to make any measurements at these center pins.**

With the coil cable disconnected, use a multimeter to assure an open circuit condition exists between the designated pin and GND, for each signal channel in Table 3-3-1.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-1

Signal Channel	Coil Cable Connector	
MC-1	A1	GND
MC-2	A15	GND
MC-3	A13	GND
MC-4	A11	GND
MC-5	A9	GND
MC-6	A7	GND
MC-7	A5	GND
MC-8	A3	GND
Transmit Signal	A20	GND
Reflected Power	A17	GND
GND	NOTE: The following points are common to GND A4, A6, A8, A10, A12, A16, A18, A22, E1 Shield, E6 Shield	

If one or more of the readings indicate a short, replace the cable. If all of the above readings indicate an open circuit condition, proceed to the **PIN Diode Control Channels Check**.

3-3-2 PIN Diode Control Channels Check

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each control channel in Table 3-3-2. Also verify that the control pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-2

Control Channel	Coil Cable Connector	System Connector
MC bias 1	A24	F2
MC bias 2	B24	F1
MC bias 3	B23	F3
MC bias 4	A23	F4
MC bias 5	B6	F5
MC bias 6	B8	F6
MC bias 7	B10	F7
MC bias 8	B4	F8
GND	A4, A6, A8, A10, A12, A16, A18, A22	G1-G8, I6, MC Shield

If one or more of the readings are > 3 ohms, replace the coil cable. If all readings are < 3 ohms, proceed to the **PIN Diodes Check**.

3-4 PIN Diodes Check

With the coil cable connected to the coil, use a multimeter to measure the diode drop across the coil connector pins as detailed in Table 3-4. Use Figure 3-3 for pin identification while performing this procedure.

PIN DIODE EXPECTED READINGS – TABLE 3-4

System Connector		Reading (volts)
Positive Lead	Negative Lead	
F1	GND [G1]	2.2 $\pm$ 0.3
GND [G1]	F1	Open
F2	GND [G2]	2.2 $\pm$ 0.3
GND [G2]	F2	Open
F3	GND [G3]	1.1 $\pm$ 0.3
GND [G3]	F3	Open
F4	GND [G4]	1.1 $\pm$ 0.3
GND [G4]	F4	Open
F5	GND [G5]	1.1 $\pm$ 0.3
GND [G5]	F5	Open
F6	GND [G6]	2.2 $\pm$ 0.3
GND [G6]	F6	Open
F7	GND [G7]	2.2 $\pm$ 0.3
GND [G7]	F7	Open
F8	GND [G8]	1.1 $\pm$ 0.3
GND [G8]	F8	Open
E1	E1 Shield	1.5 $\pm$ 0.3
E1 Shield	E1	1.7 $\pm$ 0.3
E6	E6 Shield	Open
E6 Shield	E6	Open

If one or more of the readings indicate a short, replace the coil. If all of the above readings are correct, redo 3-2 *Coil Imaging Performance Verification*. If the problem still exists, replace the 1.5T HD TR Knee Array.

3-5 Mechanical Hardware Check

Not Applicable

3-6 Troubleshooting Tips

If the following error is displayed, there may be a problem with the Coil ID coil or system hardware: *The prescribed RF coil and detected RF coil do not match.*

Troubleshooting instructions are available through the service browser or service methods CDs:

11.x/12.x: Instructions not available in service methods. Access tool via common service browser path: **Diagnostics -> Peripherals -> SRI Functional Tests -> Coil ID**

Start the diagnostic by selecting **SRI Coil ID** and **Run**. The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the trouble-shooting instructions if the correct ID is not displayed. When the trouble-shooting is complete, and if the correct ID is still not displayed, replace the cable.

HDx(14.x)/HDxT(15.x) Service Methods: **Troubleshooting -> System -> Common Service Desktop Help Screens -> SRI Coil ID – 16-channel switch**

450/450w Service Methods: **Troubleshooting -> Diagnostics -> Hardware Location -> Magnet Room -> Coil ID Functional Diagnostic**

SECTION 4 – MAINTENANCE

4-1 Coil Care



WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.



CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The 1.5T HD TR Knee Array, baseplate and pads must be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

4-2 Special Care Requirements

None

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 Disassembly of Coil

Not Applicable

5-2 External Cable Replacement



DURING EXTERNAL CABLE REPLACEMENT, THE 1.5T HD TR KNEE ARRAY COMBINER BOARD IS EXPOSED. **USE ESD SAFETY PRECAUTIONS**

5-2-1 Tools Required

- 1ea Philips Screwdriver

5-2-2 Replacement Time

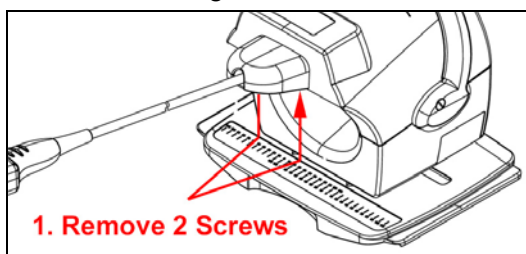
- 10 to 15 minutes

5-2-3 Replacement Procedure

The 2 coil connector mounting screws are located at the inferior end of the coil.

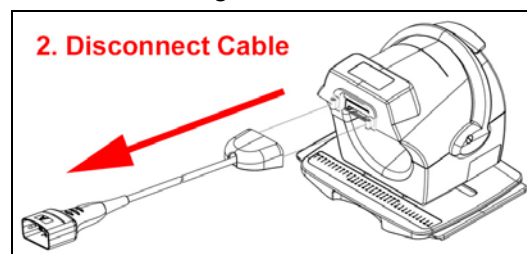
1. Remove the 2 coil connector mounting screws as detailed in Figure 5-2-3-1.
2. Remove the coil connector by grasping the coil connector housing and pulling away from the coil in the inferior direction. [Figure 5-2-3-2]
3. Using the FRU cable, reverse the above process for installation.

Figure 5-2-3-1



Remove 2 Screws

Figure 5-2-3-2



Disconnect Cable

5-3 Mechanical Hardware Replacement

5-3-1 Baseplate Replacement

5-3-1-1 Tools Required

- 1ea Philips Screwdriver

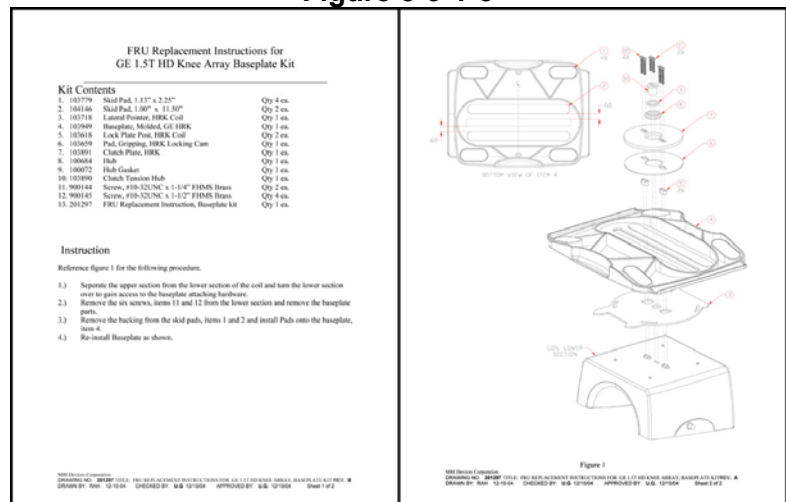
5-3-1-2 Replacement Time

- 20 to 30 minutes

5-3-1-3 Replacement Procedure

1. Open the 1.5T HD TR Knee Array Baseplate FRU kit and find the FRU Replacement Instructions for GE 1.5T HD TR Knee Array Baseplate Kit, as illustrated in Figure 5-3-1-3.
2. Follow the FRU Replacement Instructions while performing the baseplate replacement.

Figure 5-3-1-3



FRU Replacement Instructions for
GE 1.5T HD TR Knee Array Baseplate Kit

5-3-2 Latch Replacement

5-3-2-1 Tools Required

- Locking Bail Insert Removal Tool [included in kit]

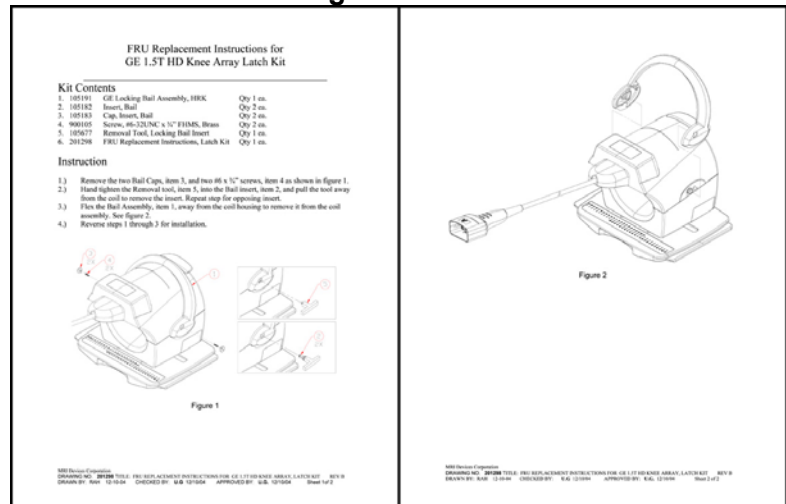
5-3-2-2 Replacement Time

- 20 to 30 minutes

5-3-2-3 Replacement Procedure

3. Open the 1.5T HD TR Knee Array Latch FRU kit and find the FRU Replacement Instructions for GE 1.5T HD TR Knee Array Latch Kit, as illustrated in Figure 5-3-2-3.
4. Follow the FRU Replacement Instructions while performing the latch replacement.

Figure 5-3-2-3



**FRU Replacement Instructions for
GE 1.5T HD TR Knee Array Latch Kit**

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	MRIDC #
1.5T HD TR Knee Array	5114356-2	870001
1.5T HD TR Knee Array Cable	5114356-3	870002
1.5T HD TR Knee Array Baseplate	5114356-4	870003
Latch Kit	5114356-13	870004
Legacy Phantom	5114356-11	870005
Phantom Positioner	5114356-12	870006
Large cylindrical unified phantom	5342679	N/A

6-2 Other Replaceable Accessories

The following table lists replaceable accessories that may be purchased from MRI Devices Corporation.

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	MRIDC #
Thin Knee Pad	870007
Thick Knee Pad	870008
Ramp Pad	870009
Ankle Pad	870010

SECTION 7 – APPENDIX

7-1 Schematic

Refer to Block Diagram under Section 1-5 Theory of Operation.

7-2 Coil Configuration

For Discovery MR450 and Optima MR450w the coil configuration is HDTRKneePA

There are 3 configurations available for HDx, HDxt, and HDe:

- HD TRKnee PA
- HDTRKneeService
- HDTRK1chService

HD TRKnee PA is the only configuration visible in clinical mode.

HDTRKneeService and **HDTRK1chService** are visible in service mode (geservice).

HDTRKneeService and **HDTRK1chService** must be installed for the SNR tool to run.

Check the system coil configuration file for these configurations. If they have not been installed, select them in the appropriate mode, from the list of available coils, and install.

Revision History

Rev	Date	Author	Primary Reason for Change
A	08-04-04	L. Hyler	Initial Release
B	11-10-04	L. Hyler	Changes per Formal Review Log, Kyle Zeller, 11.0/04
C	12-07-04	L. Hyler	QA Section Changes from Julia Klinge
D	12-21-04	L. Hyler	Add External Cable Checks from Asdrubal Mata
E	01-07-05	L. Hyler	Change Name to "1.5T HD Knee Array" Throughout Manual
F	01-13-05	L. Hyler	Implement Changes from J. Klinge e-mail of 01-13-05
01	01-21-05	L. Hyler	Released
02	6-5-08	L. Hyler	Customer Change Request CCN0021 (DCN 3994)
03	10/29/09	M. Vaughn	Updated per Change Request CCN0036 (Unified phantom set, 450/450w addition)